CY014 - CYANVision

HL7 Documentation
Host Interface Manual

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Please read the user manual before operating the instrument and make sure the installation is performed correctly.

WHO SHOULD READ THIS MANUAL

This manual is written for LIS (Laboratory Information Management System) developers and those who need to understand the working LIS principle in HL7. Instructions are provided in this manual for LIS developers to guide them to develop LIS interface that enables their LIS to communicate with the chemistry analyzer of Cypress Diagnostics. The developers are expected to have knowledge of LIS and HL7 standards, and capacity of network programming. The communication protocol is TCP/IP for network layer and HL7 version 2.3.1 for application layer. LIS developers are recommended to develop the LIS interface using Visual C++, Visual Basic, etc. in the Windows operating system.

WHAT CAN YOU BE FOUND IN THIS MANUAL

This operation manual introduces the HL7 interface used by Cypress Diagnostics on the CYANVision. Chapter 1 describes the HL7 interface generally. The CYANVision supports bidirectional communication with LIS, that is, it can not only send test results to LIS but also download sample information from LIS. Chapter 2 explains the bidirectional communication in detail.

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1 INTRODUCTION TO HL7

The HL7-standard protocol for CYANVision implemented for transmission of results to a host, external LIS and other IP-based networks. The HL7 message requires a TCP/IP connection and a software version V1.22 or higher, if not contact your distributor to perform a software upgrade. It enables the LIS host to receive test results from the biochemistry analyzers. The host interface manual described the two communication modes (sending and downloading). All data are transmitted in format of HL7 V2.3.1.

WARNING: For complete and detailed descriptions of HL7 message grammar, refer to the HL7 standard written by the HL7 standard committee. Only the essential parts – the parts need in the communication protocol are described below.

As the HL7 message does not have a predefined length, there is a frame of the following format. This way the receiver will be able to detect the different messages. These headers and trailers are usually non-printable characters and excluded from the content of (printed) HL7 messages. The prefix is a vertical tab character <VT> (hex 0x0b). The suffix is a file separator character <FS> (hex 0x1c) immediately followed by a carriage return <CR> (hex 0x0d). It will look like this:

<VT> *HL7 Message goes in here* <FS><CR>

1.0 COMPATIBLE HL7 MESSAGES

All messages used for HL7 interface include ORU, ACK and DSR. Test results are transferred as follows Sample information is downloaded from LIS as follows.

WARNING: HL7 supports many types of messages, but only 4 of them are implemented on the CYANVision analyzer.

1.1 COMPONENTS OF AN HL7 MESSAGE

Each HL7 standard message is composed of groups and segments where:

- Groups contain segments
- Segments contain fields
- Fields contain subfields

All messages used for the HL7 interface include ORU, ACK, QRY & DSR.

1.2 TRANSFER OF TEST RESULTS

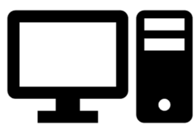
Test results are transferred as follows:

ORU^R01

>>>>>>>>>>>>>>>>>>>>>>

<<<<<<<<<<<<<<<<<<<<<<

ACK^R01



Analyzer | LIS

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1.2.1 Observe Results (unsolicited) or ORU – message

The ORU^R01 I used to transmit sample result to the LIS host, an ORU^R01 message includes: Patient information (patient name, sample ID), sample information (sample type, test, …) & test results (test NO. concentration, unit, reference, …)
A sample with multiple tests will be transmitted over multiple ORU messages.
The structure of an ORU message is as follows:

ORU – segments	Description
MSH	Message header
PID	Patient identification
OBR	Observation report ID
OBX	Observation/Result
NTE	Additional information

CYANVision will be able to include send additional data with the sample results. The additional information is included as notes (NTE) and are named “Metainformation”. It is linked to the original data by using the Sample_ID as a referral.

NTE|1|SAMPLEID_REF|Metainformation|GR|Abs|MethodID|MethodName|ProgramID|Linearity-Range

1.2.2 Acknowledgment or ACK-message

The ACK^R01 message responds to ORU messages the structure of an ACK message is as follows:

ORU – segments	Description
MSH	Message header
MSA	Message Acknowledgment

1.3 Download of worklist

Sample information is downloaded from LIS as follows:

QRY^Q02
>>>>>>>>>>>>>>>>>>>>>
DSR^Q03
<<<<<<<<<<<<<<<<<<<<<
ACK^Q03
>>>>>>>>>>>>>>>>>>>>>
DSR^Q03
<<<<<<<<<<<<<<<<<<<<<
ACK^Q03
>>>>>>>>>>>>>>>>>>>>>

Analyzer LIS

The SRD, ACK loop repeats until the final dataset is send, which is signaled by a void DSC field.

1.3.1 Query or QRY-message

QRY^Q02 message is used for sample information query on LIS and has an event Q02. The structure of QRY message is as follows:

QRY – segments	Description
MSH	Message header
QRD	Query Definition
QRF	Query Filter

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1.3.2 Display response or DSR-message

DSR^Q03 message sends and displays searched results, i.e. send sample information from LIS to the analyzer. The structure of DSR message is as follows:

DSR – segments	Description
MSH	Message header
MSA	Message Acknowledgment
ERR	Error
QAK	Query Acknowledgment
QRD	Query Definition
QRF	Query filter
DSP	Display Data
DSC	Continuation Pointer

1.4 MESSAGE SEGMENTS

WARNING: HL7 supports many types of subfields, but only a limited amount are implemented or can be interpreted by the CYANVision analyzer.

This sections described the components of each segment: field name, field length and description. All fields used in message segment are listed in following tables. The numbers followed by a '#' symbol indicate that he fields are required for the message.

If optionality indicates "O" the data is not required, if it indicates "R", the data is required.
The following abbreviations for the datatype are used (see HL7 standard for more detailed information):

Datatype abbr.	**Meaning (see standard for details)**
CE	Coded element
CQ	Composite Quantity With units
CX	Extended Composite ID with check Digit
DLN	Driver's License Number
EI	Entity Identifier
ELD	Error
EIP	Parent Order
HD	Hierarchic Designator
ID	Coded ID see section 2.6.7 of the standard
IS	Coded values for user-defined tables
MOC	Charge to Practice
MSG	Message Type
NDL	Observing Practitioner
NM	Numeric
PRL	Parent Result Link
PT	Processing Type
SI	Sequence ID
SPS	Specimen Source
ST	String
TQ	Timing Quantity
TS	Time Stamp

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TX	Text Data
VARIES	Variable Datatype
VID	Version Identifier
VR	Value Qualifier
XAD	Extended Address
XCN	Extended Composite Name and Identification Number for Organizations
XPN	Extended Person Name
XTN	Extended Telecommunication Number

1.4.1 MSH – Message header segment

All HL7 messages begin with MSH, which is the first segment of an HL7 message and always located at the beginning of the message. The MSH segment defines the intent source, destination, and some specifics of the syntax of a message.

The MSH segment is build up of the following fields:

Field	Length	Datatype	Optionality
MSH.1 - Field Separator	1	ST	R
MSH.2 - Encoding Characters	4	ST	R
MSH.3 - Sending Application	180	HD	O
MSH.4 - Sending Facility	180	HD	O
MSH.5 - Receiving Application	180	HD	O
MSH.6 - Receiving Facility	180	HD	O
MSH.7 - Date/Time Of Message	26	TS	O
MSH.8 - Security	40	ST	O
MSH.9 - Message Type	7	MSG	R
MSH.10 - Message Control ID	20	ST	R
MSH.11 - Processing ID	3	PT	R
MSH.12 - Version ID	60	VID	R
MSH.13 - Sequence Number	15	NM	O
MSH.14 - Continuation Pointer	180	ST	O
MSH.15 - Accept Acknowledgment Type	2	ID	O
MSH.16 - Application Acknowledgment Type	2	ID	O
MSH.17 - Country Code	2	ID	O
MSH.18 - Character Set	16	ID	O
MSH.19 - Principal Language Of Message	60	CE	O
MSH.20 - Alternate Character Set Handling Scheme	20	ID	O

1.4.2 MSA - Message acknowledgment segment

The MSA segment contains information sent while acknowledging another message.

The MSA segment is buildup of the following fields:

Field	Length	Datatype	Optionality
MSA.1 - Acknowledgement Code	2	ID	R
MSA.2 - Message Control ID	20	ST	R
MSA.3 - Text Message	80	ST	O
MSA.4 - Expected Sequence Number	15	NM	O

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MSA.5 - Delayed Acknowledgment Type	1	ID	B
MSA.6 - Error Condition	100	CE	O

1.4.3 ERR – Error segment

The ERR segment is used to add error comments to acknowledgment messages. It contains one field only storing the code and the location.

The ERR segment is buildup of the following fields:

Field	Length	Datatype	Optionality
ERR.1 - Error Code and Location	80	ELD	R

1.4.4 PID – Patient identification segment

The PID segment is used by all applications as the primary means of communicating patient identification information. This segment contains permanent patient identifying and demographic information that, for the most part, is not likely to change frequently.

The PID segment is buildup of the following fields:

Field	Length	Datatype	Optionality
PID.1 - Set ID - PID	4	SI	O
PID.2 - Patient ID	20	CX	B
PID.3 - Patient Identifier List	20	CX	R
PID.4 - Alternate Patient ID - PID	20	CX	B
PID.5 - Patient Name	48	XPN	R
PID.6 - Mother s Maiden Name	48	XPN	O
PID.7 - Date/Time Of Birth	26	TS	O
PID.8 - Sex	1	IS	O
PID.9 - Patient Alias	48	XPN	O
PID.10 - Race	80	CE	O
PID.11 - Patient Address	106	XAD	O
PID.12 - County Code	4	IS	B
PID.13 - Phone Number - Home	40	XTN	O
PID.14 - Phone Number - Business	40	XTN	O
PID.15 - Primary Language	60	CE	O
PID.16 - Marital Status	80	CE	O
PID.17 - Religion	80	CE	O
PID.18 - Patient Account Number	20	CX	O
PID.19 - SSN Number - Patient	16	ST	B
PID.20 - Driver's License Number - Patient	25	DLN	O
PID.21 - Mother's Identifier	20	CX	O
PID.22 - Ethnic Group	80	CE	O
PID.23 - Birth Place	60	ST	O
PID.24 - Multiple Birth Indicator	1	ID	O
PID.25 - Birth Order	2	NM	O
PID.26 - Citizenship	80	CE	O
PID.27 - Veterans Military Status	60	CE	O

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PID.28 - Nationality	80	CE	O
PID.29 - Patient Death Date and Time	26	TS	O
PID.30 - Patient Death Indicator	1	ID	O

1.4.5 OBR – Observation Request

The Observation Request (OBR) segment is used to transmit information specific to an order for a diagnostic study or observation, physical exam, or assessment.

The OBR segment is buildup of the following fields:

Field	**Length**	**Datatype**	**Optionality**
OBR.1 - Set ID - OBR	4	SI	O
OBR.2 - Placer Order Number	22	EI	C
OBR.3 - Filler Order Number	22	EI	C
OBR.4 - Universal Service ID	200	CE	R
OBR.5 - Priority-OBR	2	ID	B
OBR.6 - Requested Date/time	26	TS	B
OBR.7 - Observation Date/Time	26	TS	C
OBR.8 - Observation End Date/Time	26	TS	O
OBR.9 - Collection Volume *	20	CQ	O
OBR.10 - Collector Identifier *	60	XCN	O
OBR.11 - Specimen Action Code *	1	ID	O
OBR.12 - Danger Code	60	CE	O
OBR.13 - Relevant Clinical Info.	300	ST	O
OBR.14 - Specimen Received Date/Time *	26	TS	C
OBR.15 - Specimen Source	300	SPS	O
OBR.16 - Ordering Provider	80	XCN	O
OBR.17 - Order Callback Phone Number	40	XTN	O
OBR.18 - Placer Field 1	60	ST	O
OBR.19 - Placer Field 2	60	ST	O
OBR.20 - Filler Field 1 +	60	ST	O
OBR.21 - Filler Field 2 +	60	ST	O
OBR.22 - Results Rpt/Status Chng - Date/Time +	26	TS	C
OBR.23 - Charge to Practice +	40	MOC	O
OBR.24 - Diagnostic Serv Sect ID	10	ID	O
OBR.25 - Result Status +	1	ID	C
OBR.26 - Parent Result +	400	PRL	O
OBR.27 - Quantity/Timing	200	TQ	O
OBR.28 - Result Copies To	150	XCN	O
OBR.29 - Parent Number	200	EIP	O
OBR.30 - Transportation Mode	20	ID	O
OBR.31 - Reason for Study	300	CE	O
OBR.32 - Principal Result Interpreter +	200	NDL	O
OBR.33 - Assistant Result Interpreter +	200	NDL	O
OBR.34 - Technician +	200	NDL	O
OBR.35 - Transcriptionist +	200	NDL	O

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Field	Length	Datatype	Optionality
OBR.36 - Scheduled Date/Time +	26	TS	O
OBR.37 - Number of Sample Containers *	4	NM	O
OBR.38 - Transport Logistics of Collected Sample *	60	CE	O
OBR.39 - Collector s Comment *	200	CE	O
OBR.40 - Transport Arrangement Responsibility	60	CE	O
OBR.41 - Transport Arranged	30	ID	O
OBR.42 - Escort Required	1	ID	O
OBR.43 - Planned Patient Transport Comment	200	CE	O
OBR.44 - Procedure Code	80	CE	O
OBR.45 - Procedure Code Modifier	80	CE	O

1.4.6 OBX - Observation segment

The OBX segment is used to transmit a single observation or observation fragment. It represents the smallest indivisible unit of a report.

The OBX segment is buildup of the following fields:

Field	Length	Datatype	Optionality
OBX.1 - Set ID - OBX	4	SI	O
OBX.2 - Value Type	3	ID	C
OBX.3 - Observation Identifier	80	CE	R
OBX.4 - Observation Sub-ID	20	ST	C
OBX.5 - Observation Value	65536	VARIES	C
OBX.6 - Units	60	CE	O
OBX.7 - References Range	60	ST	O
OBX.8 - Abnormal Flags	5	ID	O
OBX.9 - Probability	5	NM	O
OBX.10 - Nature of Abnormal Test	2	ID	O
OBX.11 - Observation Result Status	1	ID	R
OBX.12 - Date Last Obs Normal Values	26	TS	O
OBX.13 - User Defined Access Checks	20	ST	O
OBX.14 - Date/Time of the Observation	26	TS	O
OBX.15 - Producer's ID	60	CE	O
OBX.16 - Responsible Observer	80	XCN	O
OBX.17 - Observation Method	60	CE	O

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1.4.7 QAK – Query Acknowledgement

The QAK segment contains information sent with responses to a query. Although the QAK segment is required in the responses to the enhanced queries (see section 2.19), it may appear as an optional segment placed after the (optional) ERR segment in any query response (message) to any original mode query.

The QAK segment is buildup of the following fields:

Field	Length	Datatype	Optionality
QAK.1 - Query Tag	32	ST	R
QAK.2 - Query Response Status	2	ID	O

1.4.8 QRD – Query definition segment

The QRD segment is used to define a query.

The QRD segment is buildup of the following fields:

Field	Length	Datatype	Optionality
QRD.1 - Query Date/Time	26	TS	R
QRD.2 - Query Format Code	1	ID	R
QRD.3 - Query Priority	1	ID	R
QRD.4 - Query ID	10	ST	R
QRD.5 - Deferred Response Type	1	ID	O
QRD.6 - Deferred Response Date/Time	26	TS	O
QRD.7 - Quantity Limited Request	10	CQ	R
QRD.8 - Who Subject Filter	60	XCN	R
QRD.9 - What Subject Filter	60	CE	R
QRD.10 - What Department Data Code	60	CE	R
QRD.11 - What Data Code Value Qual.	20	VR	O
QRD.12 - Query Results Level	1	ID	O

1.4.9 QRF – Query filter segment

The QRF segment is used with the QRD segment to further refine the content of an original style query.

The QRF segment is buildup of the following fields:

Field	Length	Datatype	Optionality
QRF.1 - Where Subject Filter	20	ST	R
QRF.2 - When Data Start Date/Time	26	TS	O
QRF.3 - When Data End Date/Time	26	TS	O
QRF.4 - What User Qualifier	60	ST	O
QRF.5 - Other QRY Subject Filter	60	ST	O
QRF.6 - Which Date/Time Qualifier	12	ID	O
QRF.7 - Which Date/Time Status Qualifier	12	ID	O
QRF.8 - Date/Time Selection Qualifier	12	ID	O
QRF.9 - When Quantity/Timing Qualifier	60	TQ	O

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1.4.10 DSP – Display data segment

The DSP segment is used to contain data that has been preformatted by the sender for display. The semantic content of the data is lost; the data is simply treated as lines of text.

The DSP segment is buildup of the following fields:

Field	Length	Datatype	Optionality
DSP.1 - Set ID - DSP	4	SI	O
DSP.2 - Display Level	4	SI	O
DSP.3 - Data Line	300	TX	R
DSP.4 - Logical Break Point	2	ST	O
DSP.5 - Result ID	20	TX	O

1.4.11 DSC – Continuation pointer segment

The DSC segment is used in the continuation protocol.

The DSC segment is buildup of the following fields:

Field	Length	Datatype	Optionality
DSC.1 - Continuation Pointer	180	ST	O

1.4.12 NTE - Notes and comments segment

The NTE segment is defined here for inclusion in messages defined in other chapters. It is a common format for sending notes and comments.

The NTE segment is buildup of the following fields:

Field	Length	Datatype	Optionality
NTE.1 - Set ID - NTE	4	SI	O
NTE.2 - Source of Comment	8	ID	O
NTE.3 - Comment	65536	FT	O
NTE.4 - Comment Type	60	CE	O

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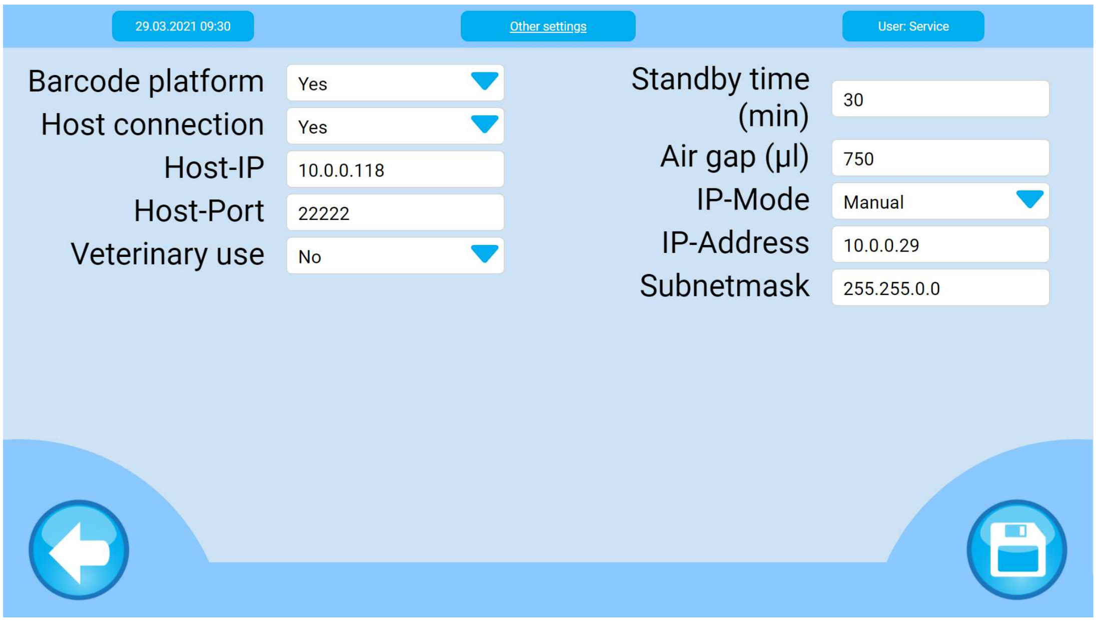
2 SETUP THE CONNECTION ON THE DEVICE

Note: The connection should only be established by a qualified service engineer, who has been authorized and trained to do so. The persons must have received an adequate training, which included the required "know-how" of IT and local network requirements.

1) Login as "Service" or "Lab Head" (Password: Cypress1).
2) Go to "Other settings", select Host-connection to "Yes".
3) Fill in the Host-IP address and the Host-port (by default this is 22222)

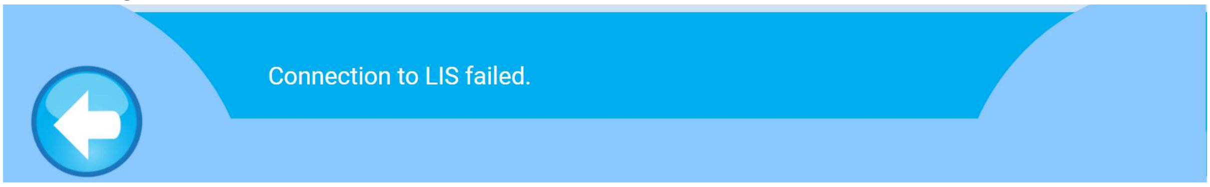
Note: Please make sure that whatever port you chose is allowed to use between the instrument and your PC (firewall rules etc.).

4) Press "Save", the host connection is now enabled on the CYANVision.



5) Test the connection
    a. Go to Patient Worklist
    b. Press the "load from LIS" button

If a warning "Connection to LIS failed" appears the connection is not established. Verify the above settings, if ok contact the IT admin of the network.



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3 COMMUNICATION PROCESS AND MESSAGE EXAMPLES

The following lines list multiple message examples of the HL7 protocol used on the CYANVision analyzer.

3.1 SENDING RESULTS FROM CYANVISION ANALYZER TO LIS HOST

The semi-automatic biochemistry analyzer – CYANVision – sends individual test result to the LIS host. The communication is triggered by pressing the "upload" button in the sample results menu.

All tests of sample are transferred by one message. The LIS host responses accordingly when receiving the message.

ORU^R01
>>>>>>>>>>>>>>>>>>>>>
<<<<<<<<<<<<<<<<<<<<<
ACK^R01

Analyzer LIS

The message type send by the analyzer is ORU R01.

ORU – segments	Description
MSH	Message header
PID	Patient identification
OBR	Observation report ID
OBX	Observation/Result
NTE	Additional information

Example of the inbound ORU^R01 message on the LIS:

```
MSH|^~\&|CYPRESS|CYANVISION|||20210318102444||ORU^R01|0000000|P|2.3.1||||0||UNICODE UTF-8|||
PID|1|123456|||Hans Dampf||20210318113332|M|||||||||||||||||||||||||||||||
OBR|1|12345|1|CypressCyanvision|Y||||||||||SERUM|||||||||||||||||||||||||||||||||||
OBX|1|NM|SampleID_REF|ProgramName|300.45|mg/L|100-200|FLAGS|||||20210319041620||||||||
NTE|1|SAMPLEID_REF|Metainformation|GR|Abs|MethodID|MethodName|ProgramID|Linearity-Range
```

The first segment or message header (MSH) looks as follows:

```
MSH|^~\&|CYPRESS|CYANVISION|||20210318102444||ORU^R01|0000000|P|2.3.1||||0||UNICODE UTF-8|||
```

Field	**Example**
MSH.1 - Field Separator	|
MSH.2 - Encoding Characters	^~\&
MSH.3 - Sending Application	CYPRESS
MSH.4 - Sending Facility	CYANVISION
MSH.5 - Receiving Application	[Empty]
MSH.6 - Receiving Facility	[Empty]

Field	**Example**
MSH.7 - Date/Time Of Message	20210318102444
MSH.8 - Security	[Empty]
MSH.9 - Message Type	ORU^R01
MSH.10 - Message Control ID	0000000
MSH.11 - Processing ID	P
MSH.12 - Version ID	2.3.1
MSH.13 - Sequence Number	[Empty]
MSH.14 - Continuation Pointer	[Empty]
MSH.15 - Accept Acknowledgment Type	[Empty]
MSH.16 - Application Acknowledgment Type	0
MSH.17 - Country Code	[Empty]
MSH.18 - Character Set	UNICODE UTF-8
MSH.19 - Principal Language Of Message	[Empty]
MSH.20 - Alternate Character Set Handling Scheme	[Empty]

Followed by the Patient identification segment (PID) look as follows:

```
PID|1|123456|||Hans Dampf||20210318113332|M|||||||||||||||||||||||||||||||
```

Field	**Example**
PID.1 - Set ID - PID	1
PID.2 - Patient ID	123456
PID.3 - Patient Identifier List	[Empty]
PID.4 - Alternate Patient ID - PID	[Empty]
PID.5 - Patient Name	Hans Dampf
PID.6 - Mother s Maiden Name	[Empty]
PID.7 - Date/Time Of Birth	20210318113332
PID.8 - Sex	M
PID.9 - Patient Alias	[Empty]
PID.10 - Race	[Empty]
PID.11 - Patient Address	[Empty]
PID.12 - County Code	[Empty]
PID.13 - Phone Number - Home	[Empty]
PID.14 - Phone Number - Business	[Empty]
PID.15 - Primary Language	[Empty]
PID.16 - Marital Status	[Empty]
PID.17 - Religion	[Empty]
PID.18 - Patient Account Number	[Empty]
PID.19 - SSN Number - Patient	[Empty]
PID.20 - Driver's License Number - Patient	[Empty]
PID.21 - Mother's Identifier	[Empty]
PID.22 - Ethnic Group	[Empty]
PID.23 - Birth Place	[Empty]
PID.24 - Multiple Birth Indicator	[Empty]
PID.25 - Birth Order	[Empty]

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PID.26 - Citizenship	[Empty]
PID.27 - Veterans Military Status	[Empty]
PID.28 - Nationality	[Empty]
PID.29 - Patient Death Date and Time	[Empty]
PID.30 - Patient Death Indicator	[Empty]

Followed by the Observation report ID segment (OBR) look as follows:

```
OBR|1|12345|1|CypressCyanvision|Y||||||||||SERUM|||||||||||||||||||||||||||||||||||
```

Field	Example
OBR.1 - Set ID - OBR	1
OBR.2 - Placer Order Number	12345
OBR.3 - Filler Order Number	1
OBR.4 - Universal Service ID	CypressCyanvision
OBR.5 - Priority-OBR	Y
OBR.6 - Requested Date/time	[EMPTY]
OBR.7 - Observation Date/Time	[EMPTY]
OBR.8 - Observation End Date/Time	[EMPTY]
OBR.9 - Collection Volume *	[EMPTY]
OBR.10 - Collector Identifier *	[EMPTY]
OBR.11 - Specimen Action Code *	[EMPTY]
OBR.12 - Danger Code	[EMPTY]
OBR.13 - Relevant Clinical Info.	[EMPTY]
OBR.14 - Specimen Received Date/Time *	[EMPTY]
OBR.15 - Specimen Source	SERUM
OBR.16 - Ordering Provider	[EMPTY]
OBR.17 - Order Callback Phone Number	[EMPTY]
OBR.18 - Placer Field 1	[EMPTY]
OBR.19 - Placer Field 2	[EMPTY]
OBR.20 - Filler Field 1 +	[EMPTY]
OBR.21 - Filler Field 2 +	[EMPTY]
OBR.22 - Results Rpt/Status Chng - Date/Time +	[EMPTY]
OBR.23 - Charge to Practice +	[EMPTY]
OBR.24 - Diagnostic Serv Sect ID	[EMPTY]
OBR.25 - Result Status +	[EMPTY]
OBR.26 - Parent Result +	[EMPTY]
OBR.27 - Quantity/Timing	[EMPTY]
OBR.28 - Result Copies To	[EMPTY]
OBR.29 - Parent Number	[EMPTY]
OBR.30 - Transportation Mode	[EMPTY]
OBR.31 - Reason for Study	[EMPTY]
OBR.32 - Principal Result Interpreter +	[EMPTY]
OBR.33 - Assistant Result Interpreter +	[EMPTY]
OBR.34 - Technician +	[EMPTY]
OBR.35 - Transcriptionist +	[EMPTY]
OBR.36 - Scheduled Date/Time +	[EMPTY]

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Field	Example
OBR.37 - Number of Sample Containers *	[EMPTY]
OBR.38 - Transport Logistics of Collected Sample *	[EMPTY]
OBR.39 - Collector s Comment *	[EMPTY]
OBR.40 - Transport Arrangement Responsibility	[EMPTY]
OBR.41 - Transport Arranged	[EMPTY]
OBR.42 - Escort Required	[EMPTY]
OBR.43 - Planned Patient Transport Comment	[EMPTY]
OBR.44 - Procedure Code	[EMPTY]
OBR.45 - Procedure Code Modifier	[EMPTY]

Followed by the Observation/Result segment (OBX) look as follows:

```
OBX|1|NM|SampleID_REF|ProgramName|300.45|mg/L|100-200|FLAGS||||||20210319041620||||||||
```

Field	Example
OBX.1 - Set ID - OBX	1
OBX.2 - Value Type	NM
OBX.3 - Observation Identifier	SampleID_REF
OBX.4 - Observation Sub-ID	ProgramName
OBX.5 - Observation Value	300.45
OBX.6 - Units	mg/L
OBX.7 - References Range	100-200
OBX.8 - Abnormal Flags	Flags
OBX.9 - Probability	[EMPTY]
OBX.10 - Nature of Abnormal Test	[EMPTY]
OBX.11 - Observation Result Status	[EMPTY]
OBX.12 - Date Last Obs Normal Values	[EMPTY]
OBX.13 - User Defined Access Checks	[EMPTY]
OBX.14 - Date/Time of the Observation	20210319041620
OBX.15 - Producer's ID	[EMPTY]
OBX.16 - Responsible Observer	[EMPTY]
OBX.17 - Observation Method	[EMPTY]

Followed by the Additional information (NTE) look as follows:

```
NTE|1|SAMPLEID_REF|Metainformation|GR|Abs|MethodID|MethodName|ProgramID|Linearity-Range
```

Field	Example
NTE.1 - Set ID - NTE	1
NTE.2 - Source of Comment	SAMPLEID_REF
NTE.3 - Comment	Metainformation
NTE.4 - Comment Type	GR
NTE.5	Abs
NTE.6	MethodID

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Field	Example
NTE.7	MethodName
NTE.8	ProgramID
NTE.9	Linearity-Range

3.2 Send a worklist from the LIS host to the CYANVision

The semi-automatic biochemistry analyzer – CYANVision – can process a worklist created and send by the LIS host. The communication is triggered by pressing the “download” button in the patient worklist menu.

When requesting that order, the LIS host responses with sending the data to the analyzer.

QRY^Q02
>>>>>>>>>>>>>>>>>>>>>
DSR^Q03
<<<<<<<<<<<<<<<<<<<<<
ACK^Q03
>>>>>>>>>>>>>>>>>>>>>
DSR^Q03
<<<<<<<<<<<<<<<<<<<<<
ACK^Q03
>>>>>>>>>>>>>>>>>>>>>

Analyzer LIS

The message type send by the analyzer is DSR^Q03

DSR – segments	Description
MSH	Message header
MSA	Message Acknowledgment
ERR	Error
QAK	Query Acknowledgment
QRD	Query Definition
QRF	Query filter
DSP	Display Data
DSC	Continuation Pointer

Example of the inbound DSR^Q03 message on the LIS:

```
MSH|^~\&|||Manufacturer|Model|20070723170000||DSR^Q03|1|P|2.3.1|||||ASCII|||
MSA|AA|1|Message                                                    accepted|||0|
ERR|0|
QAK|SR|OK|
QRD|20070723170000|R|D|1|||RD||OTH|||T|
QRF|CyanVision|20070723000000|20070723170000|||RCT|COR|ALL||
DSP|1||JD123|||
DSP|2||Y|||
DSP|3||Johnathana|||
DSP|4||Does|||
DSP|5||F|||
DSP|6||19550604000000|||
DSP|7||1|||
DSP|8||GLUC|||
DSC||
```


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The first segment or message header (MSH) looks as follows:

MSH|^~\&|||Manufacturer|Model|20070723170000||DSR^Q03|1|P|2.3.1||||||ASCII|||

Field	Example
MSH.1 - Field Separator	|
MSH.2 - Encoding Characters	^~\&
MSH.3 - Sending Application	[Empty]
MSH.4 - Sending Facility	[Empty]
MSH.5 - Receiving Application	Manufacturer
MSH.6 - Receiving Facility	Model
MSH.7 - Date/Time Of Message	20070723170000
MSH.8 - Security	[Empty]
MSH.9 - Message Type	DSR^Q03
MSH.10 - Message Control ID	1
MSH.11 - Processing ID	P
MSH.12 - Version ID	2.3.1
MSH.13 - Sequence Number	[Empty]
MSH.14 - Continuation Pointer	[Empty]
MSH.15 - Accept Acknowledgment Type	[Empty]
MSH.16 - Application Acknowledgment Type	0
MSH.17 - Country Code	[Empty]
MSH.18 - Character Set	ASCII
MSH.19 - Principal Language Of Message	[Empty]
MSH.20 - Alternate Character Set Handling Scheme	[Empty]

The message acknowledgement segment (MSA) looks as follows:

MSA|AA|1|Message accepted|||0|

Field	Example
MSA.1 - Acknowledgement Code	AA
MSA.2 - Message Control ID	1
MSA.3 - Text Message	Message accepted
MSA.4 - Expected Sequence Number	[Empty]
MSA.5 - Delayed Acknowledgment Type	[Empty]
MSA.6 - Error Condition	0

The error segment (ERR) looks as follows:

ERR|0|

Field	Example
ERR.1 - Error Code and Location	0

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The Query Acknowledgment segment (QAK) looks as follows:

QAK|SR|OK|

Field	**Example**
QAK.1 - Query Tag	SR
QAK.2 - Query Response Status	OK

The Query Definition segment (QRD) looks as follows:

QRD|20070723170000|R|D|1|||RD||OTH|||T|

Field	**Example**
QRD.1 - Query Date/Time	20070723170000
QRD.2 - Query Format Code	R
QRD.3 - Query Priority	D
QRD.4 - Query ID	1
QRD.5 - Deferred Response Type	[Empty]
QRD.6 - Deferred Response Date/Time	[Empty]
QRD.7 - Quantity Limited Request	RD
QRD.8 - Who Subject Filter	[Empty]
QRD.9 - What Subject Filter	OTH
QRD.10 - What Department Data Code	[Empty]
QRD.11 - What Data Code Value Qual.	[Empty]
QRD.12 - Query Results Level	T

The Query Filter segment (QRF) look as follows:

QRF|CyanVision|20070723000000|20070723170000|||RCT|COR|ALL||

Field	**Example**
QRF.1 - Where Subject Filter	CyanVision
QRF.2 - When Data Start Date/Time	20070723000000
QRF.3 - When Data End Date/Time	20070723170000
QRF.4 - What User Qualifier	[Empty]
QRF.5 - Other QRY Subject Filter	[Empty]
QRF.6 - Which Date/Time Qualifier	RCT
QRF.7 - Which Date/Time Status Qualifier	COR
QRF.8 - Date/Time Selection Qualifier	ALL
QRF.9 - When Quantity/Timing Qualifier	[Empty]

Multiple DSP segments can follow each other. For the example it looks as follows:

```
DSP|1||JD123|||
DSP|2||Y|||
DSP|3||Johnathana|||
DSP|4||Does|||
DSP|5||F|||
DSP|6||19550604000000|||
```


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DSP|7||1||| **DSP|8||GLUC|||**

Field	Examples							
DSP.1 - Set ID - DSP	1	2	3	4	5	6	7	8
DSP.2 - Display Level	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]
DSP.3 - Data Line	JD123	Y	Johnathana	DOES	F	19550604000000	1	GLUC
DSP.4 - Logical Break Point	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]
DSP.5 - Result ID	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]	[Empty]

The communication is finished with a continuation pointer segment (DSC), which look as follows:

DSC||

Field	Example
DSC.1 - Continuation Pointer	[Empty]

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